lab25(SSU)

December 3, 2025

1 Lab 25: Linear Regression

Helpful Resource:

- Python Reference: Cheat sheet of helpful array & table methods!

Recommended Readings:

- Correlation
- The Regression Line
- Method of Least Squares
- Least Squares Regression

[1]:
```
!apt-get install texlive texlive-xetex texlive-latex-extra pandoc
!pip install pypandoc
!pip install datascience
```

```
Reading package lists… Done
Building dependency tree… Done
Reading state information… Done
pandoc is already the newest version (2.9.2.1-3ubuntu2).
pandoc set to manually installed.
The following additional packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
  fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
  libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
  libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
  libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzzip-0-13
  lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
  ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
  teckit tex-common tex-gyre texlive-base texlive-binaries
  texlive-fonts-recommended texlive-latex-base texlive-latex-recommended
  texlive-pictures texlive-plain-generic tipa xfonts-encodings xfonts-utils
Suggested packages:
  fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
  libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
  poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
  fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
  fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
```

```
  | postscript-viewer perl-tk xpdf | pdf-viewer xzdec
  texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
  icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
  texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
  texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
  default-jre-headless tipa-doc
The following NEW packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
  fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
  libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
  libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
  libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzzip-0-13
  lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
  ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
  teckit tex-common tex-gyre texlive texlive-base texlive-binaries
  texlive-fonts-recommended texlive-latex-base texlive-latex-extra
  texlive-latex-recommended texlive-pictures texlive-plain-generic
  texlive-xetex tipa xfonts-encodings xfonts-utils
0 upgraded, 54 newly installed, 0 to remove and 41 not upgraded.
Need to get 182 MB of archives.
After this operation, 571 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-droid-fallback all
1:6.0.1r16-1.1build1 [1,805 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-lato all 2.0-2.1
[2,696 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 poppler-data all
0.4.11-1 [2,171 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-common all 6.17
[33.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-urw-base35 all
20200910-1 [6,367 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libidn12 amd64
1.38-4ubuntu1 [60.0 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy/main amd64 libijs-0.35 amd64
0.35-15build2 [16.5 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/main amd64 libjbig2dec0 amd64
0.19-3build2 [64.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libkpathsea6
amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/main amd64 libwoff1 amd64
1.0.2-1build4 [45.2 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/universe amd64 dvisvgm amd64
2.13.1-1 [1,221 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-lmodern all
```

```
2.004.5-6.1 [4,532 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-noto-mono all
20201225-1build1 [397 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-texgyre all
20180621-3.1 [10.2 MB]
Get:17 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libapache-pom-java
all 18-1 [4,720 B]
Get:18 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-parent-
java all 43-1 [10.8 kB]
Get:19 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-logging-
java all 1.2-2 [60.3 kB]
Get:20 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libptexenc1
amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
Get:21 http://archive.ubuntu.com/ubuntu jammy/main amd64 rubygems-integration
all 1.18 [5,336 B]
Get:22 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby3.0 amd64
3.0.2-7ubuntu2.11 [50.1 kB]
Get:23 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-rubygems
all 3.3.5-2ubuntu1.2 [228 kB]
Get:24 http://archive.ubuntu.com/ubuntu jammy/main amd64 ruby amd64 1:3.0~exp1
[5,100 B]
Get:25 http://archive.ubuntu.com/ubuntu jammy/main amd64 rake all 13.0.6-2 [61.7
kB]
Get:26 http://archive.ubuntu.com/ubuntu jammy/main amd64 ruby-net-telnet all
0.1.1-2 [12.6 kB]
Get:27 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-webrick
all 1.7.0-3ubuntu0.2 [52.5 kB]
Get:28 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-xmlrpc all
0.3.2-1ubuntu0.1 [24.9 kB]
Get:29 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libruby3.0
amd64 3.0.2-7ubuntu2.11 [5,114 kB]
Get:30 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libsynctex2
amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
Get:31 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libteckit0 amd64
2.5.11+ds1-1 [421 kB]
Get:32 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtexlua53
amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
Get:33 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtexluajit2
amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
Get:34 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libzzip-0-13 amd64
0.13.72+dfsg.1-1.1 [27.0 kB]
Get:35 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-encodings all
1:1.0.5-0ubuntu2 [578 kB]
Get:36 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-utils amd64
1:7.7+6build2 [94.6 kB]
Get:37 http://archive.ubuntu.com/ubuntu jammy/universe amd64 lmodern all
2.004.5-6.1 [9,471 kB]
Get:38 http://archive.ubuntu.com/ubuntu jammy/universe amd64 preview-latex-style
```

```
all 12.2-1ubuntu1 [185 kB]
Get:39 http://archive.ubuntu.com/ubuntu jammy/main amd64 t1utils amd64
1.41-4build2 [61.3 kB]
Get:40 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:42 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
Get:43 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive all
2021.20220204-1 [14.3 kB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 4s (51.1 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages …
Selecting previously unselected package fonts-droid-fallback.
(Reading database … 121713 files and directories currently installed.)
Preparing to unpack …/00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
…
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) …
Selecting previously unselected package fonts-lato.
Preparing to unpack …/01-fonts-lato_2.0-2.1_all.deb …
Unpacking fonts-lato (2.0-2.1) …
Selecting previously unselected package poppler-data.
Preparing to unpack …/02-poppler-data_0.4.11-1_all.deb …
Unpacking poppler-data (0.4.11-1) …
Selecting previously unselected package tex-common.
```

```
Preparing to unpack …/03-tex-common_6.17_all.deb …
Unpacking tex-common (6.17) …
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack …/04-fonts-urw-base35_20200910-1_all.deb …
Unpacking fonts-urw-base35 (20200910-1) …
Selecting previously unselected package libgs9-common.
Preparing to unpack …/05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb …
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) …
Selecting previously unselected package libidn12:amd64.
Preparing to unpack …/06-libidn12_1.38-4ubuntu1_amd64.deb …
Unpacking libidn12:amd64 (1.38-4ubuntu1) …
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack …/07-libijs-0.35_0.35-15build2_amd64.deb …
Unpacking libijs-0.35:amd64 (0.35-15build2) …
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack …/08-libjbig2dec0_0.19-3build2_amd64.deb …
Unpacking libjbig2dec0:amd64 (0.19-3build2) …
Selecting previously unselected package libgs9:amd64.
Preparing to unpack …/09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb …
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) …
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack …/10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
…
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) …
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack …/11-libwoff1_1.0.2-1build4_amd64.deb …
Unpacking libwoff1:amd64 (1.0.2-1build4) …
Selecting previously unselected package dvisvgm.
Preparing to unpack …/12-dvisvgm_2.13.1-1_amd64.deb …
Unpacking dvisvgm (2.13.1-1) …
Selecting previously unselected package fonts-lmodern.
Preparing to unpack …/13-fonts-lmodern_2.004.5-6.1_all.deb …
Unpacking fonts-lmodern (2.004.5-6.1) …
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack …/14-fonts-noto-mono_20201225-1build1_all.deb …
Unpacking fonts-noto-mono (20201225-1build1) …
Selecting previously unselected package fonts-texgyre.
Preparing to unpack …/15-fonts-texgyre_20180621-3.1_all.deb …
Unpacking fonts-texgyre (20180621-3.1) …
Selecting previously unselected package libapache-pom-java.
Preparing to unpack …/16-libapache-pom-java_18-1_all.deb …
Unpacking libapache-pom-java (18-1) …
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack …/17-libcommons-parent-java_43-1_all.deb …
Unpacking libcommons-parent-java (43-1) …
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack …/18-libcommons-logging-java_1.2-2_all.deb …
Unpacking libcommons-logging-java (1.2-2) …
```

```
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack …/19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
…
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) …
Selecting previously unselected package rubygems-integration.
Preparing to unpack …/20-rubygems-integration_1.18_all.deb …
Unpacking rubygems-integration (1.18) …
Selecting previously unselected package ruby3.0.
Preparing to unpack …/21-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb …
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) …
Selecting previously unselected package ruby-rubygems.
Preparing to unpack …/22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb …
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) …
Selecting previously unselected package ruby.
Preparing to unpack …/23-ruby_1%3a3.0~exp1_amd64.deb …
Unpacking ruby (1:3.0~exp1) …
Selecting previously unselected package rake.
Preparing to unpack …/24-rake_13.0.6-2_all.deb …
Unpacking rake (13.0.6-2) …
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack …/25-ruby-net-telnet_0.1.1-2_all.deb …
Unpacking ruby-net-telnet (0.1.1-2) …
Selecting previously unselected package ruby-webrick.
Preparing to unpack …/26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb …
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) …
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack …/27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb …
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) …
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack …/28-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb …
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) …
Selecting previously unselected package libsynctex2:amd64.
Preparing to unpack …/29-libsynctex2_2021.20210626.59705-1ubuntu0.2_amd64.deb
…
Unpacking libsynctex2:amd64 (2021.20210626.59705-1ubuntu0.2) …
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack …/30-libteckit0_2.5.11+ds1-1_amd64.deb …
Unpacking libteckit0:amd64 (2.5.11+ds1-1) …
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack …/31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
…
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) …
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
…/32-libtexluajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb …
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) …
Selecting previously unselected package libzzip-0-13:amd64.
Preparing to unpack …/33-libzzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb …
```

```
Unpacking libzzip-0-13:amd64 (0.13.72+dfsg.1-1.1) …
Selecting previously unselected package xfonts-encodings.
Preparing to unpack …/34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb …
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) …
Selecting previously unselected package xfonts-utils.
Preparing to unpack …/35-xfonts-utils_1%3a7.7+6build2_amd64.deb …
Unpacking xfonts-utils (1:7.7+6build2) …
Selecting previously unselected package lmodern.
Preparing to unpack …/36-lmodern_2.004.5-6.1_all.deb …
Unpacking lmodern (2.004.5-6.1) …
Selecting previously unselected package preview-latex-style.
Preparing to unpack …/37-preview-latex-style_12.2-1ubuntu1_all.deb …
Unpacking preview-latex-style (12.2-1ubuntu1) …
Selecting previously unselected package t1utils.
Preparing to unpack …/38-t1utils_1.41-4build2_amd64.deb …
Unpacking t1utils (1.41-4build2) …
Selecting previously unselected package teckit.
Preparing to unpack …/39-teckit_2.5.11+ds1-1_amd64.deb …
Unpacking teckit (2.5.11+ds1-1) …
Selecting previously unselected package tex-gyre.
Preparing to unpack …/40-tex-gyre_20180621-3.1_all.deb …
Unpacking tex-gyre (20180621-3.1) …
Selecting previously unselected package texlive-binaries.
Preparing to unpack …/41-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb …
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) …
Selecting previously unselected package texlive-base.
Preparing to unpack …/42-texlive-base_2021.20220204-1_all.deb …
Unpacking texlive-base (2021.20220204-1) …
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack …/43-texlive-fonts-recommended_2021.20220204-1_all.deb …
Unpacking texlive-fonts-recommended (2021.20220204-1) …
Selecting previously unselected package texlive-latex-base.
Preparing to unpack …/44-texlive-latex-base_2021.20220204-1_all.deb …
Unpacking texlive-latex-base (2021.20220204-1) …
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack …/45-texlive-latex-recommended_2021.20220204-1_all.deb …
Unpacking texlive-latex-recommended (2021.20220204-1) …
Selecting previously unselected package texlive.
Preparing to unpack …/46-texlive_2021.20220204-1_all.deb …
Unpacking texlive (2021.20220204-1) …
Selecting previously unselected package libfontbox-java.
Preparing to unpack …/47-libfontbox-java_1%3a1.8.16-2_all.deb …
Unpacking libfontbox-java (1:1.8.16-2) …
Selecting previously unselected package libpdfbox-java.
Preparing to unpack …/48-libpdfbox-java_1%3a1.8.16-2_all.deb …
Unpacking libpdfbox-java (1:1.8.16-2) …
Selecting previously unselected package texlive-pictures.
```

```
Preparing to unpack …/49-texlive-pictures_2021.20220204-1_all.deb …
Unpacking texlive-pictures (2021.20220204-1) …
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack …/50-texlive-latex-extra_2021.20220204-1_all.deb …
Unpacking texlive-latex-extra (2021.20220204-1) …
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack …/51-texlive-plain-generic_2021.20220204-1_all.deb …
Unpacking texlive-plain-generic (2021.20220204-1) …
Selecting previously unselected package tipa.
Preparing to unpack …/52-tipa_2%3a1.3-21_all.deb …
Unpacking tipa (2:1.3-21) …
Selecting previously unselected package texlive-xetex.
Preparing to unpack …/53-texlive-xetex_2021.20220204-1_all.deb …
Unpacking texlive-xetex (2021.20220204-1) …
Setting up fonts-lato (2.0-2.1) …
Setting up fonts-noto-mono (20201225-1build1) …
Setting up libwoff1:amd64 (1.0.2-1build4) …
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) …
Setting up libijs-0.35:amd64 (0.35-15build2) …
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) …
Setting up libfontbox-java (1:1.8.16-2) …
Setting up rubygems-integration (1.18) …
Setting up libzzip-0-13:amd64 (0.13.72+dfsg.1-1.1) …
Setting up fonts-urw-base35 (20200910-1) …
Setting up poppler-data (0.4.11-1) …
Setting up tex-common (6.17) …
update-language: texlive-base not installed and configured, doing nothing!
Setting up libjbig2dec0:amd64 (0.19-3build2) …
Setting up libteckit0:amd64 (2.5.11+ds1-1) …
Setting up libapache-pom-java (18-1) …
Setting up ruby-net-telnet (0.1.1-2) …
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) …
Setting up t1utils (1.41-4build2) …
Setting up libidn12:amd64 (1.38-4ubuntu1) …
Setting up fonts-texgyre (20180621-3.1) …
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) …
Setting up ruby-webrick (1.7.0-3ubuntu0.2) …
Setting up fonts-lmodern (2.004.5-6.1) …
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) …
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) …
Setting up libsynctex2:amd64 (2021.20210626.59705-1ubuntu0.2) …
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) …
Setting up teckit (2.5.11+ds1-1) …
Setting up libpdfbox-java (1:1.8.16-2) …
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) …
Setting up preview-latex-style (12.2-1ubuntu1) …
Setting up libcommons-parent-java (43-1) …
Setting up dvisvgm (2.13.1-1) …
```

```
Setting up libcommons-logging-java (1.2-2) …
Setting up xfonts-utils (1:7.7+6build2) …
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) …
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) …
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) …
Setting up texlive-base (2021.20220204-1) …
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST…
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN…
mktexlsr: Updating /var/lib/texmf/ls-R…
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) …
Setting up texlive-plain-generic (2021.20220204-1) …
Setting up texlive-latex-base (2021.20220204-1) …
Setting up texlive-latex-recommended (2021.20220204-1) …
Setting up texlive-pictures (2021.20220204-1) …
Setting up texlive-fonts-recommended (2021.20220204-1) …
Setting up tipa (2:1.3-21) …
Setting up texlive (2021.20220204-1) …
Setting up texlive-latex-extra (2021.20220204-1) …
Setting up texlive-xetex (2021.20220204-1) …
Setting up rake (13.0.6-2) …
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) …
Setting up ruby3.0 (3.0.2-7ubuntu2.11) …
Setting up ruby (1:3.0~exp1) …
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) …
Processing triggers for man-db (2.10.2-1) …
Processing triggers for mailcap (3.70+nmu1ubuntu1) …
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) …
Processing triggers for libc-bin (2.35-0ubuntu3.8) …
/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link
```

```
/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

Processing triggers for tex-common (6.17) …
Running updmap-sys. This may take some time… done.
Running mktexlsr /var/lib/texmf … done.
Building format(s) --all.
        This may take some time… done.
Collecting pypandoc
  Downloading pypandoc-1.16.2-py3-none-any.whl.metadata (16 kB)
Downloading pypandoc-1.16.2-py3-none-any.whl (19 kB)
Installing collected packages: pypandoc
Successfully installed pypandoc-1.16.2
Collecting datascience
  Downloading datascience-0.18.1-py3-none-any.whl.metadata (910 bytes)
Requirement already satisfied: folium>=0.9.1 in /usr/local/lib/python3.12/dist-
packages (from datascience) (0.20.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-
packages (from datascience) (75.2.0)
Requirement already satisfied: matplotlib>=3.0.0 in
/usr/local/lib/python3.12/dist-packages (from datascience) (3.10.0)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
(from datascience) (2.2.2)
```

```
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from datascience) (1.16.3)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from datascience) (2.0.2)
Requirement already satisfied: ipython in /usr/local/lib/python3.12/dist-
packages (from datascience) (7.34.0)
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages
(from datascience) (5.24.1)
Requirement already satisfied: branca in /usr/local/lib/python3.12/dist-packages
(from datascience) (0.8.2)
Requirement already satisfied: jinja2>=2.9 in /usr/local/lib/python3.12/dist-
packages (from folium>=0.9.1->datascience) (3.1.6)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-
packages (from folium>=0.9.1->datascience) (2.32.4)
Requirement already satisfied: xyzservices in /usr/local/lib/python3.12/dist-
packages (from folium>=0.9.1->datascience) (2025.11.0)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
(1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.0.0->datascience) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
(4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
(1.4.9)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
(25.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.0.0->datascience) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
(3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
(2.9.0.post0)
Collecting jedi>=0.16 (from ipython->datascience)
  Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)
Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (4.4.2)
Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (0.7.5)
Requirement already satisfied: traitlets>=4.2 in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (5.7.1)
Requirement already satisfied: prompt-toolkit!=3.0.0,!=3.0.1,<3.1.0,>=2.0.0 in
/usr/local/lib/python3.12/dist-packages (from ipython->datascience) (3.0.52)
```

```
Requirement already satisfied: pygments in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (2.19.2)
Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (0.2.0)
Requirement already satisfied: matplotlib-inline in
/usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.2.1)
Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (4.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tenacity>=6.2.0 in
/usr/local/lib/python3.12/dist-packages (from plotly->datascience) (9.1.2)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in
/usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython->datascience)
(0.8.5)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from
jinja2>=2.9->folium>=0.9.1->datascience) (3.0.3)
Requirement already satisfied: ptyprocess>=0.5 in
/usr/local/lib/python3.12/dist-packages (from pexpect>4.3->ipython->datascience)
(0.7.0)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-
packages (from prompt-
toolkit!=3.0.0,!=3.0.1,<3.1.0,>=2.0.0->ipython->datascience) (0.2.14)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib>=3.0.0->datascience) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->folium>=0.9.1->datascience) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2025.11.12)
Downloading datascience-0.18.1-py3-none-any.whl (725 kB)
                          725.1/725.1 kB
42.5 MB/s eta 0:00:00
Downloading jedi-0.19.2-py2.py3-none-any.whl (1.6 MB)
                          1.6/1.6 MB
70.6 MB/s eta 0:00:00
Installing collected packages: jedi, datascience
Successfully installed datascience-0.18.1 jedi-0.19.2
```

[2]:
```
# Connect Google Drive to Colab so you can access your files
from google.colab import drive
drive.mount('/content/drive')

import os
os.chdir('/content/drive/MyDrive/Colab Notebooks/')
```

```
Mounted at /content/drive
```

[3]:
```
# Run this cell to set up the notebook, but please don't change it.

import numpy as np
from datascience import *

# These lines do some fancy plotting magic.
import matplotlib
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
import warnings
warnings.simplefilter('ignore', FutureWarning)
from datetime import datetime
```

1.1 1. Linear Regression Setup

When performing linear regression, we need to compute several important quantities which will be used throughout our analysis. **Unless otherwise specified when asked to make a prediction please assume we are predicting y from x throughout this assignment.** To help with our later analysis, we will begin by writing some of these functions and understanding what they can do for us.

Question 1.1. Define a function `standard_units` that converts a given array to standard units. **(3 points)**

Hint: You may find the `np.mean` and `np.std` functions helpful.

[4]:
```
def standard_units(data):
    return (data - np.mean(data)) / np.std(data)
```

Question 1.2. Which of the following are true about standard units? Assume we have converted an array of data into standard units using the function above. **(5 points)**

1. The unit of all our data when converted into standard units is the same as the unit of the original data.
2. The sum of all our data when converted into standard units is 0.
3. The standard deviation of all our data when converted into standard units is 1.
4. Adding a constant, C, to our original data has no impact on the resultant data when converted to standard units.

5. Multiplying our original data by a positive constant, C (>0), has no impact on the resultant data when converted to standard units.

Assign `standard_array` to an array of your selections, in increasing numerical order. For example, if you wanted to select options 1, 3, and 5, you would assign `standard_array` to `make_array(1, 3, 5)`.

[5]:
```
standard_array = make_array(2, 3, 4, 5)
```

Question 1.3. Define a function `correlation` that computes the correlation between 2 arrays of data in original units. **(3 points)**

Hint: Feel free to use functions you have defined previously.

[6]:
```
def correlation(x, y):
    x_su = standard_units(x)
    y_su = standard_units(y)
    return np.mean(x_su * y_su)
```

Question 1.4. Which of the following are true about the correlation coefficient r? **(5 points)**

1. The correlation coefficient measures the strength of a linear relationship.
2. When looking at the existing data, a correlation coefficient of 1.0 means an increase in one variable always means an increase in the other variable.
3. The correlation coefficient is the slope of the regression line in standard units.
4. The correlation coefficient stays the same if we swap our x-axis and y-axis.
5. If we add a constant, C, to our original data, our correlation coefficient will increase by the same C.

Assign `r_array` to an array of your selections, in increasing numerical order. For example, if you wanted to select options 1, 3, and 5, you would assign `r_array` to `make_array(1, 3, 5)`.

[7]:
```
r_array = make_array(1, 2, 3, 4)
```

Question 1.5. Define a function `slope` that computes the slope of our line of best fit (to predict y given x), given two arrays of data in original units. Assume we want to create a line of best fit in original units. **(3 points)**

Hint: Feel free to use functions you have defined previously.

[8]:
```
def slope(x, y):
    r = correlation(x, y)
    return r * (np.std(y) / np.std(x))
```

Question 1.6. Which of the following are true about the slope of our line of best fit? Assume `x` refers to the value of one variable that we use to predict the value of `y`. **(5 points)**

1. In original units, the slope has the unit: unit of x / unit of y.
2. In standard units, the slope is unitless.
3. In original units, the slope is unchanged by swapping x and y.
4. In standard units, a slope of 1 means our data is perfectly linearly correlated.
5. In original units and standard units, the slope always has the same positive or negative sign.

Assign `slope_array` to an array of your selections, in increasing numerical order. For example, if you wanted to select options 1, 3, and 5, you would assign `slope_array` to `make_array(1, 3, 5)`.

[9]:
```
slope_array = make_array(2, 4, 5)
```

Question 1.7. Define a function `intercept` that computes the intercept of our line of best fit (to predict y given x), given 2 arrays of data in original units. Assume we want to create a line of best fit in original units. **(3 points)**

Hint: Feel free to use functions you have defined previously.

[10]:
```
def intercept(x, y):
    return np.mean(y) - slope(x, y) * np.mean(x)
```

Question 1.8. Which of the following are true about the intercept of our line of best fit? Assume `x` refers to the value of one variable that we use to predict the value of `y`. **(5 points)**

1. In original units, the intercept has the same unit as the y values.
2. In original units, the intercept has the same unit as the x values.
3. In original units, the slope and intercept have the same unit.
4. In standard units, the intercept for the regression line is 0.
5. In original units and standard units, the intercept always has the same numerical value.

Assign `intercept_array` to an array of your selections, in increasing numerical order. For example, if you wanted to select options 1, 3, and 5, you would assign `intercept_array` to `make_array(1, 3, 5)`.

[11]:
```
intercept_array = make_array(1, 4)
```

Question 1.9. Define a function `predict` that takes in a table and 2 column names, and returns an array of predictions. The predictions should be created using a fitted **regression line**. We are predicting `"col2"` from `"col1"`, both in original units. **(5 points)**

Hint 1: Feel free to use functions you have defined previously.

Hint 2: Re-reading 15.2 might be helpful here.

Note: The public tests are quite comprehensive for this question, so passing them means that your function most likely works correctly.

[12]:
```
def predict(tbl, col1, col2):
    x = tbl.column(col1)
    y = tbl.column(col2)
    a = slope(x, y)
    b = intercept(x, y)
    return a * x + b
```

1.2 2. FIFA Predictions

The following data was scraped from sofifa.com, a website dedicated to collecting information from FIFA video games. The dataset consists of all players in FIFA 22 and their corresponding attributes. We have truncated the dataset to a limited number of rows (100) to ease with our visualizations and analysis. Since we're learning about linear regression, we will look specifically for a linear association between various player attributes. **To help with understanding where the line of best fit generated in linear regression comes from please do not use the `.fit_line` argument in `.scatter` at any point on question 2 unless the code was provided for you.**

Feel free to read more about the video game on Wikipedia.

[13]:
```
# Run this cell to load the data
fifa = Table.read_table('./DS/fifa22.csv')

fifa = fifa.take(np.arange(100))
# Select a subset of columns to analyze (there are 110 columns in the original␣
↪dataset)
fifa = fifa.select("short_name", "overall", "value_eur", "wage_eur", "age",␣
↪"pace", "shooting", "passing", "attacking_finishing")
fifa.show(5)
```

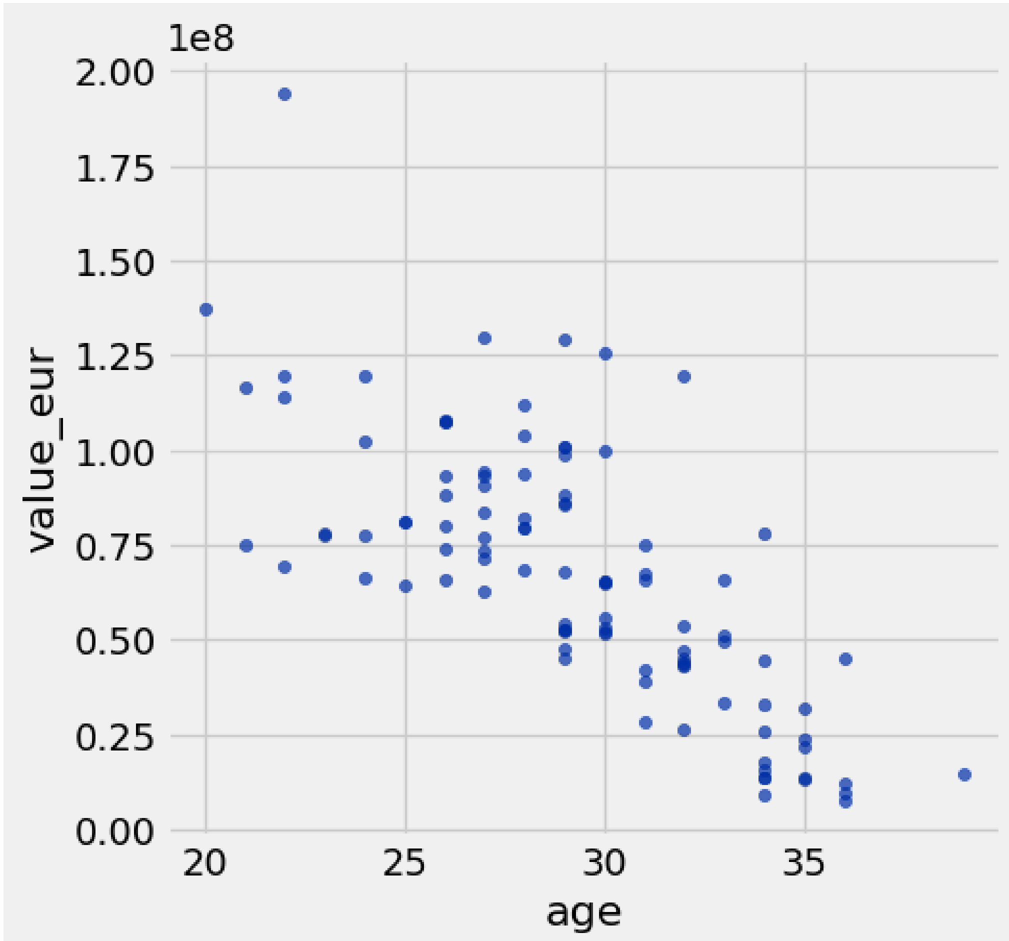
```
/usr/local/lib/python3.12/dist-packages/datascience/tables.py:163: DtypeWarning:
Columns (25,108) have mixed types. Specify dtype option on import or set
low_memory=False.
  df = pandas.read_csv(filepath_or_buffer, *args, **vargs)
```

```
<IPython.core.display.HTML object>
```

Question 2.1. Before jumping into any statistical techniques, it's important to see what the data looks like, because data visualizations allow us to uncover patterns in our data that would have otherwise been much more difficult to see. **(3 points)**

Create a scatter plot with age on the x-axis ("age"), and the player's value in Euros ("value_eur") on the y-axis.

[14]:
```
fifa.scatter('age', 'value_eur')
```



Question 2.2. Does the correlation coefficient `r` for the data in our scatter plot in 2.1 look closest to 0, 0.75, or -0.75? **(3 points)**

Assign `r_guess` to one of 0, 0.75, or -0.75.

[15]:
```
r_guess = 0.75
```

Question 2.3. Create a scatter plot with player age ("age") along the x-axis and both real player value ("value_eur") and predicted player value along the y-axis. The predictions should be created using a fitted **regression line**. The color of the dots for the real player values should be different from the color for the predicted player values. **(8 points)**

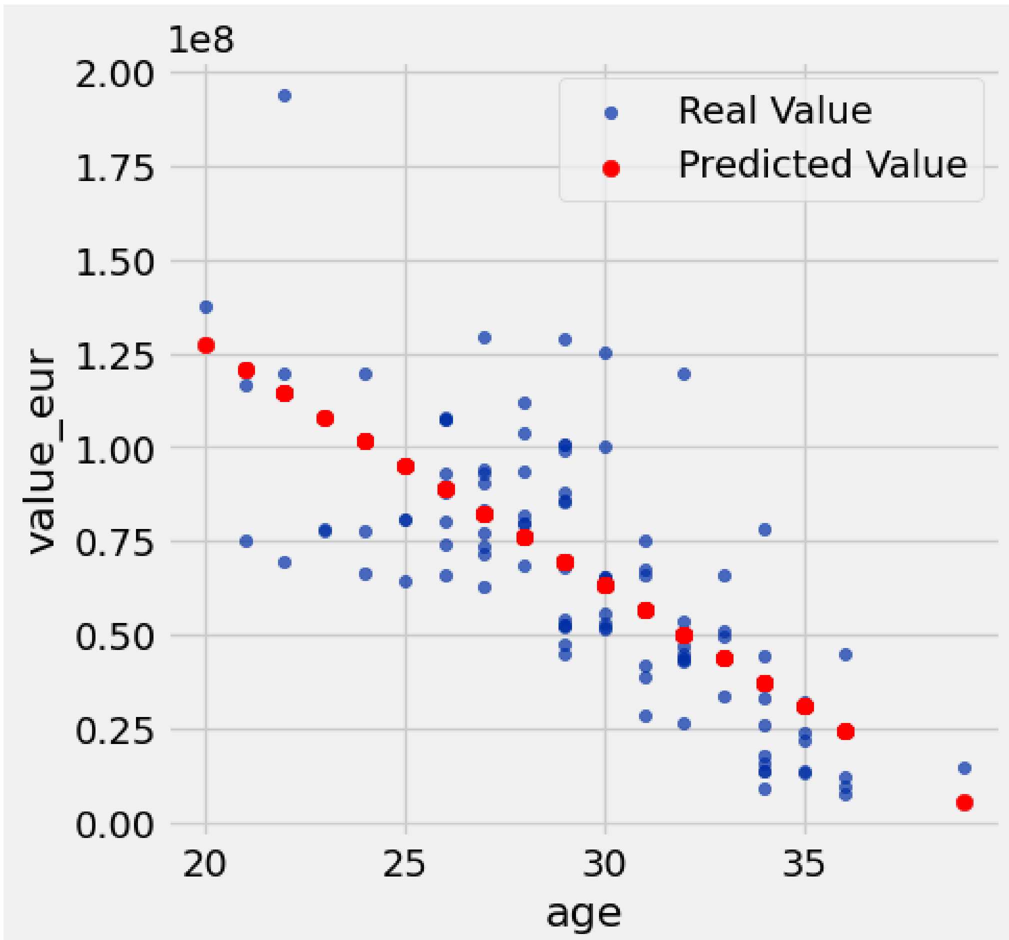
Hint 1: Feel free to use functions you have defined previously.

Hint 2: 15.2 and 7.3 has examples of creating such scatter plots.

[16]:
```
predictions = predict(fifa, 'age', 'value_eur')
fifa_with_predictions = fifa.with_columns(
    'Predicted Value', predictions
)
```

```
fifa_with_predictions.scatter('age', 'value_eur', label='Real Value')
plt.scatter(fifa_with_predictions.column('age'), fifa_with_predictions.
 ↪column('Predicted Value'), color='red', label='Predicted Value')
plt.legend()
```

[16]: <matplotlib.legend.Legend at 0x7d154d3fa8a0>



Question 2.4. Looking at the scatter plot you produced above, is linear regression a good model to use? If so, what features or characteristics make this model reasonable? If not, what features or characteristics make it unreasonable? **(5 points)**

Linear regression may not be a good model for this data. Looking at the scatterplot, player value initially increases with age, peaks around age 25-30, and then declines. This is closer to a U-shaped or parabolic relationship. Because linear regression assumes a straight line, it fails to capture the curvilinear pattern in this data and can lead to significant discrepancies between predicted and actual values. Therefore, it may tend to underestimate or overestimate the actual value for both young and older players.

Question 2.5. In 2.3, we created a scatter plot **in original units**. Now, create a scatter plot with player age **in standard units** along the x-axis and both real and predicted player value **in standard units** along the y-axis. The color of the dots of the real and predicted values should be different. **(8 points)**

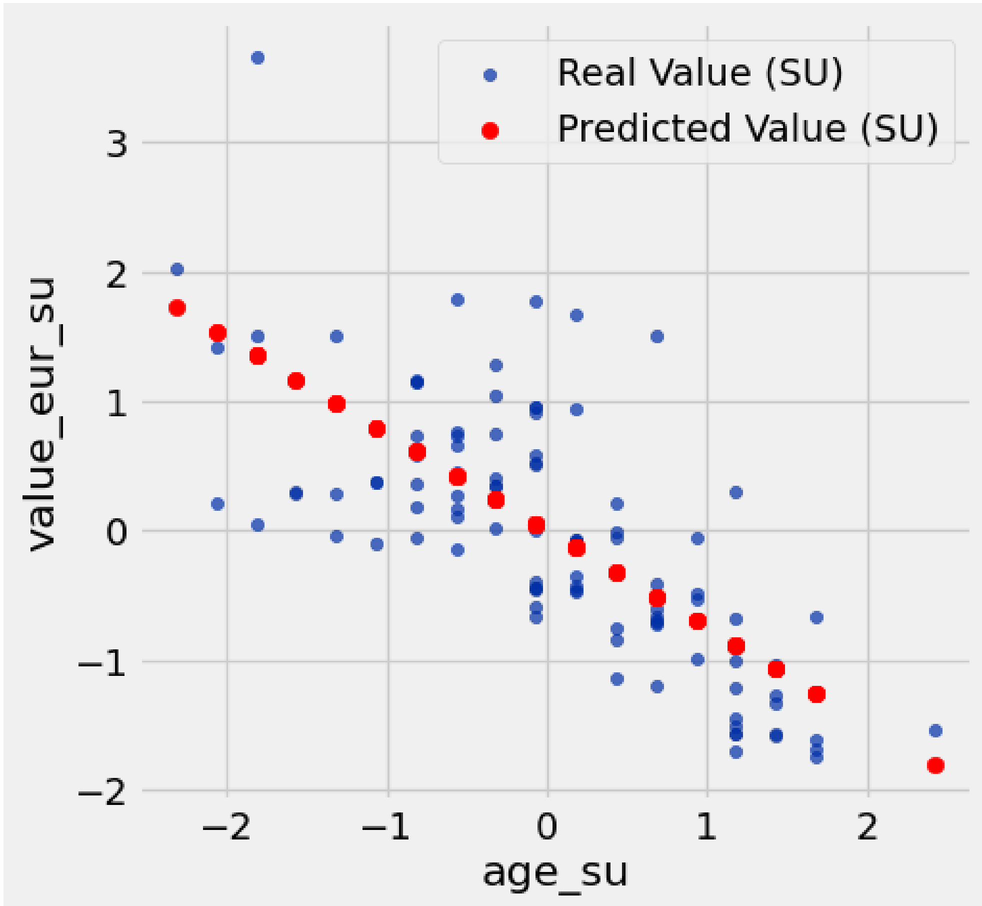
Hint: Feel free to use functions you have defined previously.

[17]:

```
age_su = standard_units(fifa.column('age'))
value_eur_su = standard_units(fifa.column('value_eur'))

fifa_su_table = Table().with_columns(
    'age_su', age_su,
    'value_eur_su', value_eur_su
)

predictions_su = predict(fifa_su_table, 'age_su', 'value_eur_su')

fifa_with_predictions_su = fifa_su_table.with_columns(
    'Predicted Value (SU)', predictions_su
)

fifa_with_predictions_su.scatter('age_su', 'value_eur_su', label='Real Value (SU)')
plt.scatter(fifa_with_predictions_su.column('age_su'), fifa_with_predictions_su.column('Predicted Value (SU)'), color='red', label='Predicted Value (SU)')
plt.legend()
```

[17]:

```
<matplotlib.legend.Legend at 0x7d154d405bb0>
```



Question 2.6. Compare your plots in 2.3 and 2.5. What similarities do they share? What differences do they have? **(5 points)**

Similarities: The two graphs share similar patterns in terms of shape and relative distribution of points. That is, the trend of increasing values with age, peaking at a certain point, and then decreasing is observed in both graphs. This is because standardization transformation only adjusts the scale without changing the underlying relationship between the data.

Differences: The key difference is the scale and units of the axes. The graph in Question 2.3 displays the data in its original units (years for age and euros for value), while the graph in Question 2.5 displays the data in standard units (mean 0, standard deviation 1). This causes the graph in Question 2.5 to be centered around 0 on both the x- and y-axes, and the absolute values of the points are smaller than those in the graph in Question 2.3. However, the relative positions and patterns of the points remain the same.

Question 2.7. Define a function `rmse` that takes in two arguments: a slope and an intercept for a potential regression line. The function should return the root mean squared error between the values predicted by a regression line with the given slope and intercept and the actual outcomes. **(6 points)**

Assume we are still predicting "value_eur" from "age" in original units from the `fifa` table.

[18]:
```
def rmse(slope, intercept):
    x = fifa.column('age')
    y = fifa.column('value_eur')
    predictions = slope * x + intercept
    errors = y - predictions
    return np.sqrt(np.mean(errors ** 2))
```

Question 2.8. Use the `rmse` function you defined along with `minimize` to find the least-squares regression parameters predicting player value from player age. Here's an example of using the `minimize` function from the textbook. **(10 points)**

Then set `lsq_slope` and `lsq_intercept` to be the least-squares regression line slope and intercept, respectively.

Finally, create a scatter plot like you did in 2.3 with player age ("age") along the x-axis and both real player value ("value_eur") and predicted player value along the y-axis. **Be sure to use your least-squares regression line to compute the predicted values.** The color of the dots for the real player values should be different from the color for the predicted player values.

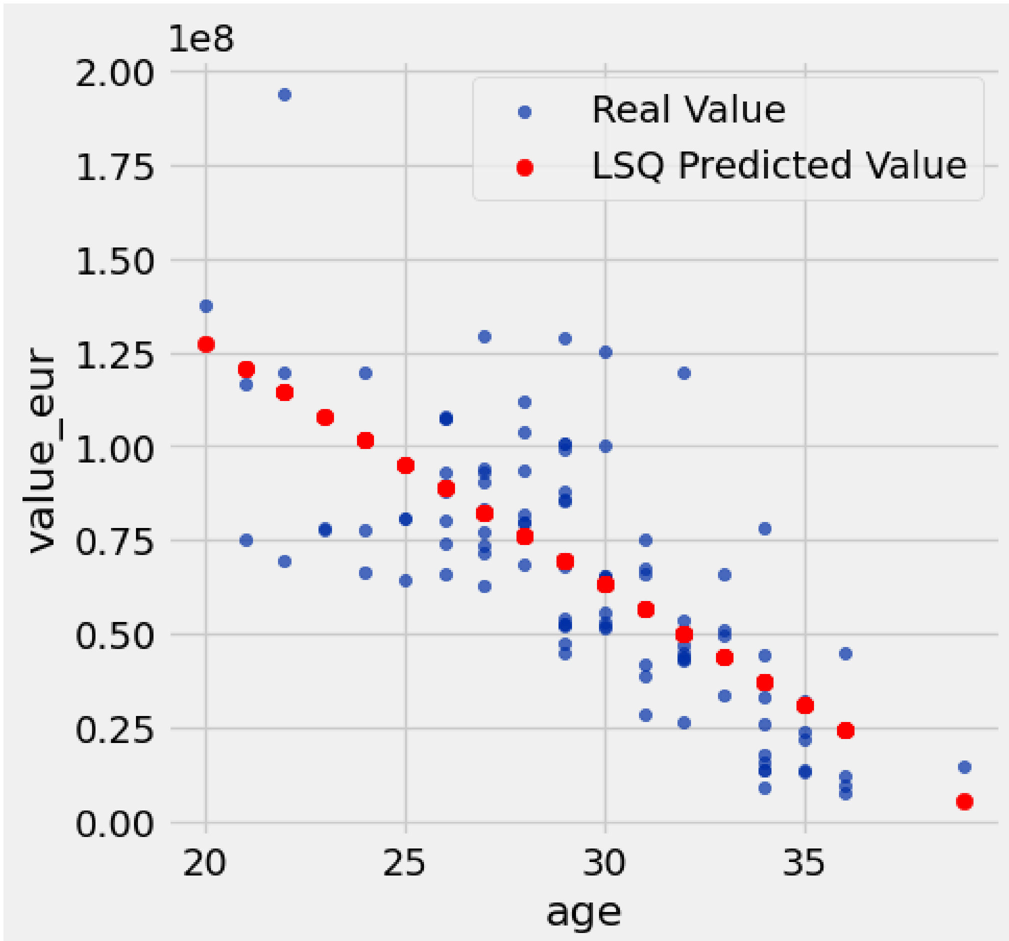
> **Note:** Your solution should not make any calls to the slope or intercept functions defined earlier.

Hint: Your call to `minimize` will return an array of argument values that minimize the return value of the function passed to `minimize`.

[19]:
```
from datascience.predicates import are

minimized_parameters = minimize(rmse)
lsq_slope = minimized_parameters.item(0)
lsq_intercept = minimized_parameters.item(1)

# This just prints your slope and intercept
print("Slope: {:g} | Intercept: {:g}".format(lsq_slope, lsq_intercept))

# Calculate predictions using the least-squares line
x_values = fifa.column('age')
lsq_predictions = lsq_slope * x_values + lsq_intercept

fifa_with_lsq_predictions = fifa.with_columns(
    'Predicted Value (LSQ)', lsq_predictions
)

fifa_with_lsq_predictions.scatter('age', 'value_eur', label='Real Value')
plt.scatter(fifa_with_lsq_predictions.column('age'), fifa_with_lsq_predictions.
  ↪column('Predicted Value (LSQ)'), color='red', label='LSQ Predicted Value')
plt.legend()
```

```
Slope: -6.41462e+06 | Intercept: 2.55525e+08
```

[19]: <matplotlib.legend.Legend at 0x7d154abe5580>



Question 2.9. The resulting line you found in 2.8 should appear very similar to the line you found in 2.3. Why were we able to minimize RMSE to find nearly the same slope and intercept from the previous formulas? **(5 points)**

Hint: Re-reading 15.3 might be helpful here.

Minimizing the Root Mean Squared Error (RMSE) is the core principle of the Least Squares Method. Least Squares finds a regression line by minimizing the sum of the squares of the vertical distances (residuals) between the observed data and the regression line. Since RMSE is the square root of the average of these squared residuals, minimizing RMSE is mathematically equivalent to minimizing the sum (or mean) of the squared residuals.

Therefore, the slope and intercept functions we defined in Question 2.3 directly compute the solution to this Least Squares Method. In other words, they are mathematical formulas for the slope and intercept that minimize the sum of squared residuals using the correlation coefficient, standard deviation, and mean. In contrast, Question 2.8 uses the minimize function to find the slope and intercept that "numerically" minimize the RMSE function. Since both methods share the same goal of "minimizing the sum of squared residuals," theoretically, they should find the same optimal line (the least squares regression line). Although there may be very slight differences due to some floating point errors in the numerical optimization process, the results are essentially the same.

Question 2.10 For which of the following error functions would we have resulted in the same slope and intercept values in 2.8 instead of using RMSE? Assume `error` is assigned to the actual values minus the predicted values. **(5 points)**

1. `np.sum(error) ** 0.5`
2. `np.sum(error ** 2)`
3. `np.mean(error) ** 0.5`
4. `np.mean(error ** 2)`

Assign `error_array` to an array of your selections, in increasing numerical order. For example, if you wanted to select options 1, 3, and 5, you would assign `error_array` to `make_array(1, 3, 5)`.

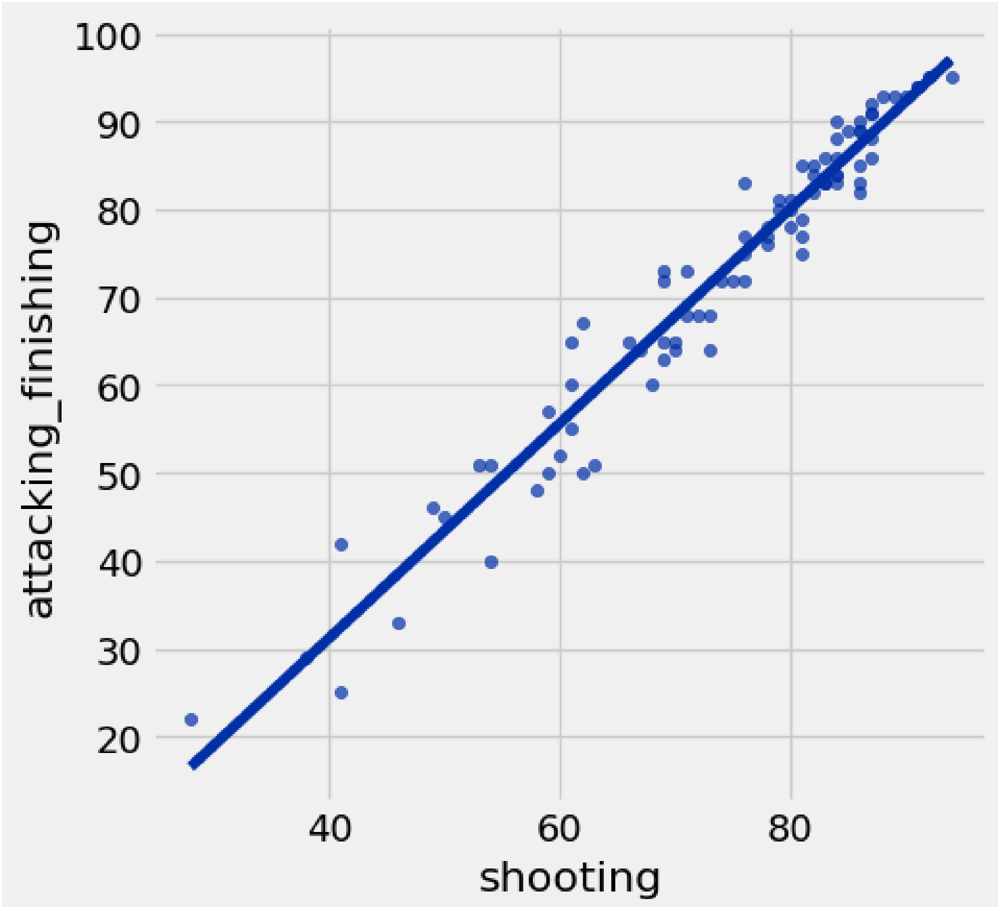
Hint: What was the purpose of RMSE? Are there any alternatives, and if so, does minimizing them them yield the same results as minimizing the RMSE?

[20]:
```
error_array = make_array(2, 4)
```

[21]:
```
# goalies don't have shooting in our dataset so we removed them before looking␣
↪at the pace stat
no_goalies = fifa.where("shooting", are.above(0))
no_goalies
```

[21]:
```
short_name         | overall | value_eur | wage_eur | age  | pace | shooting |
passing | attacking_finishing
L. Messi           | 93      | 7.8e+07   | 320000   | 34   | 85   | 92       | 91
| 95
R. Lewandowski     | 92      | 1.195e+08 | 270000   | 32   | 78   | 92       | 79
| 95
Cristiano Ronaldo  | 91      | 4.5e+07   | 270000   | 36   | 87   | 94       | 80
| 95
Neymar Jr          | 91      | 1.29e+08  | 270000   | 29   | 91   | 83       | 86
| 83
K. De Bruyne       | 91      | 1.255e+08 | 350000   | 30   | 76   | 86       | 93
| 82
K. Mbappé          | 91      | 1.94e+08  | 230000   | 22   | 97   | 88       | 80
| 93
H. Kane            | 90      | 1.295e+08 | 240000   | 27   | 70   | 91       | 83
| 94
N. Kanté           | 90      | 1e+08     | 230000   | 30   | 78   | 66       | 75
| 65
K. Benzema         | 89      | 6.6e+07   | 350000   | 33   | 76   | 86       | 81
| 90
H. Son             | 89      | 1.04e+08  | 220000   | 28   | 88   | 87       | 82
| 88
… (75 rows omitted)
```

[22]:
```python
# Run this cell to generate a scatter plot for the next part.
no_goalies.scatter('shooting', 'attacking_finishing', fit_line=True)
```



Question 2.11. Above is a scatter plot showing the relationship between a player's shooting ability ("shooting") and their scoring ability ("attacking_finishing").

There is clearly a strong positive correlation between the 2 variables, and we'd like to predict a player's scoring ability from their shooting ability. Which of the following are true, assuming linear regression is a reasonable model? **(5 points)**

Hint: Re-reading 15.2 might be helpful here.

1. For a majority of players with a `shooting` attribute above 80 our model predicts they have a better scoring ability than shooting ability.
2. A randomly selected player's predicted scoring ability in standard units will always be less than their shooting ability in standard units.
3. If we select a player who's shooting ability is 1.0 in standard units, their scoring ability, on average, will be less than 1.0 in standard units.
4. Goalies have attacking_finishing scores in our dataset but do not have shooting scores. We can still use our model to predict their attacking_finishing scores.

Assign `scoring_array` to an array of your selections, in increasing numerical order. For example, if you wanted to select options 1, 3, and 5, you would assign `scoring_array` to `make_array(1, 3, 5)`.

```
[23]: scoring_array = make_array(3)
```

1.3 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a pdf file for you to submit. **Please save before exporting!**

```
[24]: # should change the directory and file name matching to yours
      !jupyter nbconvert './DS/lab25(SSU).ipynb' --to pdf
```

```
[NbConvertApp] Converting notebook ./DS/lab25(SSU).ipynb to pdf
[NbConvertApp] Writing 87602 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 92966 bytes to DS/lab25(SSU).pdf
```